

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Business Problem Statement

A leading retail company seeks to better understand customer shopping behavior in order to improve sales performance, customer engagement, and long-term customer retention. The company has collected transactional data containing customer demographics, purchasing habits, product preferences, subscription status, seasonal shopping trends, and customer reviews.

To support data-driven decision-making, the management team requires an analysis that identifies purchasing patterns, customer segments, and factors influencing shopping behavior.

This project analyzes the customer shopping behavior dataset using Python for data preparation and exploratory analysis, PostgreSQL for structured business queries, and Power BI for dashboard development. The insights generated from this analysis will enable stakeholders to improve marketing strategies, optimize product offerings, strengthen customer engagement, and support business growth.

Deliverables

The project consists of the following deliverables:

1. Data Preparation, Cleaning and Modelling (Python)

Prepared the dataset by inspecting its structure, handling missing values, standardizing column names, engineering new features, and ensuring the data was suitable for analysis.

2. Database Integration and SQL Analysis (PostgreSQL)

Loaded the cleaned dataset into PostgreSQL and executed SQL queries to answer key business questions and extract actionable insights.

3. Interactive Dashboard (Power BI)

Developed an interactive dashboard to visualize important business metrics, customer behavior, and purchasing trends.

4. Project Documentation

Documented the complete analytical process, including data preparation, exploratory analysis, SQL findings, dashboard visualizations, business insights, and recommendations.

5. GitHub Repository

Organized all project files, including the Jupyter Notebook, SQL scripts, Power BI dashboard, and project documentation in a structured GitHub repository.

Project Overview

This project examines customer shopping behavior using a retail transaction dataset containing customer demographic information, purchasing activities, product preferences, subscription status, seasonal trends, and customer review ratings. The objective of the analysis is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

The findings from this project provide valuable information that can assist stakeholders in understanding customer purchasing behavior and support more informed marketing and product decisions.

Dataset Summary

- **Dataset Name:** Customer Shopping Behavior Dataset
- **Rows:** 3,900
- **Columns (Original):**18
- **Dataset Shape:**(3900, 18)
- **Key Features**
 - Customer demographics (Customer ID, Age, Gender, Location)
 - Product information (Item Purchased, Category, Size, Color)
 - Purchase details (Purchase Amount, Previous Purchases, Frequency of Purchases)
 - Customer engagement (Subscription Status, Discount Applied, Promo Code Used)
 - Shopping behavior (Season, Shipping Type, Payment Method, Review Rating)
- **Missing Data:** Initial inspection showed 37 missing values in the Review Rating column. These missing values were handled during the data cleaning stage using the median review rating within each product category.

Dataset Structure

```
<Class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                   3900 non-null   int64
2   Gender                               3900 non-null   str
3   Item Purchased                       3900 non-null   str
4   Category                             3900 non-null   str
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   str
7   Size                                  3900 non-null   str
8   Color                                 3900 non-null   str
9   Season                               3900 non-null   str
10  Review Rating                        3863 non-null   float64
11  Subscription Status                  3900 non-null   str
12  Shipping Type                       3900 non-null   str
13  Discount Applied                     3900 non-null   str
14  Promo Code Used                      3900 non-null   str
15  Previous Purchases                   3900 non-null   int64
16  Payment Method                       3900 non-null   str
17  Frequency of Purchases               3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

Figure 1: Output of *df.info()*

Summary Statistics

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

Figure 2: Output of *df.describe()*

Sample Dataset

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

Figure 3: Output of `df.head()`

DATA CLEANING and Exploratory Data Analysis (EDA)

- **Missing Values:** The inspection revealed that only the **Review Rating** column contained missing values, with **37 missing records**, while all other variables were complete. To preserve the dataset and avoid unnecessary data loss, the missing review ratings were imputed using the **median review rating within each product category**. This approach maintains the central tendency of customer ratings while accounting for differences across product categories.
- **Duplicate Records:** There were no duplicate records in the dataset
- **Column Standardization:** column names were standardized using the **snake_case** naming convention to improve readability and simplify coding.
- **Redundant Variable Check:** A consistency check was performed to determine whether the **Discount Applied** and **Promo Code Used** columns contained identical information, the **Promo Code Used** column was removed
- **Feature Engineering:**
 - Labels column were created for the different age groups
 - Created `purchase_frequency_days` column from the `frequency_of_purchase` column
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

DATA ANALYSIS USING SQL

The cleaned data was loaded into PostgreSQL for structured analysis to answer key business questions

- **Revenue Generated by Gender:** Calculated the total purchase amount generated by male and female customers

	gender text	sum numeric
1	Female	75191
2	Male	157890

Figure 4: Revenue by Gender Query Result

- **High Spending Discount User:** To identify high-value customers who remain active even when promotions are offered.

	customer_id	discount_applied	purchase_amount
	text	bigint	
1	2	Yes	64
2	3	Yes	73
3	4	Yes	90
4	7	Yes	85
5	9	Yes	97
6	12	Yes	68
7	13	Yes	72
8	16	Yes	81
9	20	Yes	90
10	22	Yes	62

Figure 5: High-Spending Customers Using Discounts

- **Top Five Products by Rating:** Identify the highest-rated products.

	item_purchased	review
	text	numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Figure 6: Highest Rated Products

- **Average Purchase Amount by Shipping Type:** Compared the average purchase amount between Standard and Express shipping

	shipping_type text	average_purchase_amount numeric
1	Standard	58.46
2	Express	60.48

Figure 7: Average Purchase Amount by Shipping Type

- **Spending Behaviour of Subscribers and Non-Subscribers:** Evaluates the purchasing behaviour of subscribed and non-subscribed customers using average spend, total customer and total revenue

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	No	2847	59.87	170436.00
2	Yes	1053	59.49	62645.00

Figure 8: Subscription Status Analysis

- **Discount Dependent Product:** identify the five products with the highest discount usage.

	item_purchased text	total_purchased numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

Figure 9: Products Most Frequently Purchased with Discounts

- **Customer Segmentation Based on Previous Purchases:** Classify customers into three groups based on their purchase history.
 1. New: 1 previous purchase
 2. Returning: 2–10 previous purchases
 3. Loyal: More than 10 previous purchases

	customer_segment text	Number of Customer bigint
1	Loyal	3116
2	New	83
3	Returning	701

Figure 10: Customer Segmentation

- **Top Three Most Purchased Products in Each Category:** The top three products in every category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Figure 11: Top Products by Category

- **Subscription Status of Repeat Buyers:** Identified customers with more than five previous purchases and grouped them according to their subscription status to determine whether repeat buyers are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Figure 12: Repeat Buyers and Subscription Status

- **Revenue Contribution by Age Group:** The total revenue generated by each age group and ranked the groups according to their revenue contribution.

	age_group text	total_revenue numeric
1	Adult	87576
2	Middle Aged	67711
3	Senior	43164
4	Young Adult	34630

Figure 13: Revenue Contribution by Age Group

POWER BI DASHBOARD

The dashboard combines key performance indicators (KPIs) with interactive charts and filters, enabling users to explore purchasing patterns, customer demographics, product performance, and spending trends.

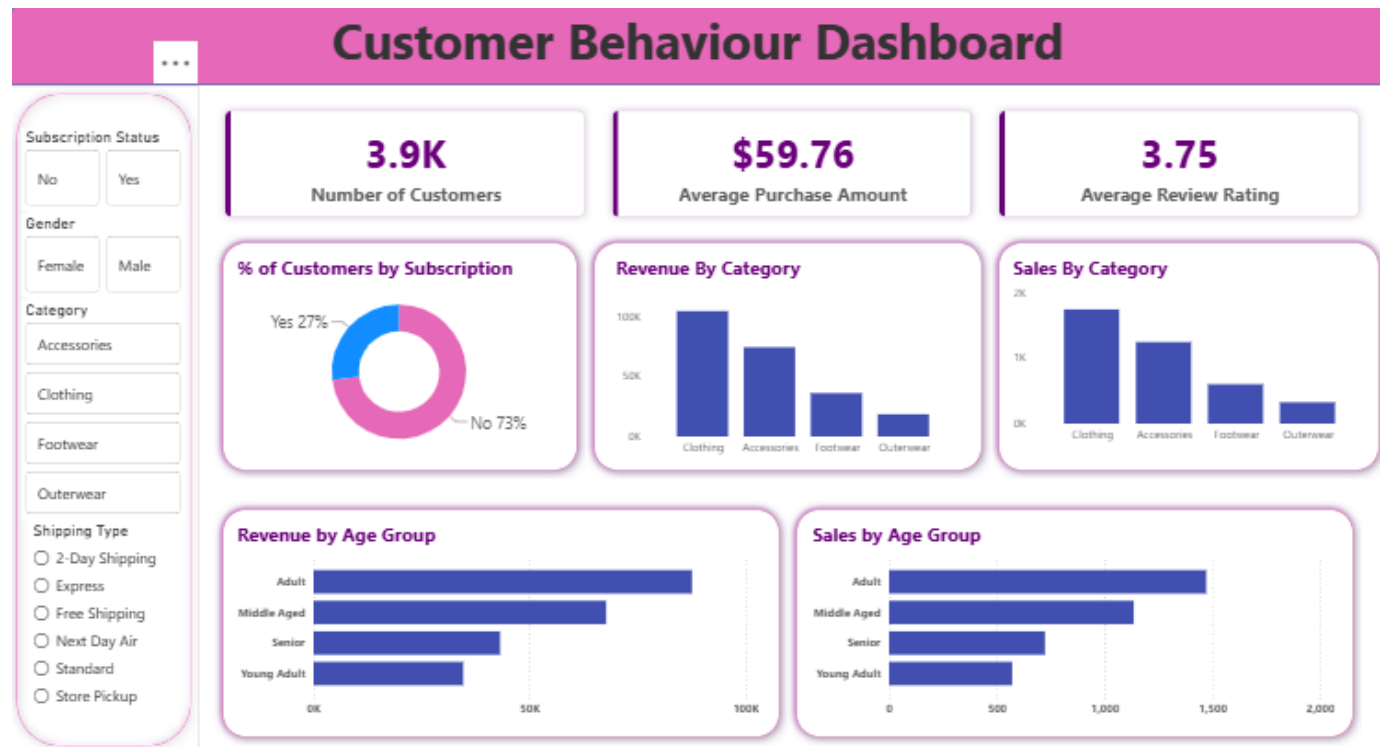


Figure 14: Customer Shopping Behavior Dashboard

KEY FINDINGS

Based on the Python analysis, SQL queries, and Power BI dashboard, the following key insights were identified:

- The **Clothing category** has the highest revenue of **\$104,262**, while **Outer wear category** has the lowest revenue of **\$18,524**. Clothing category generated a **45% of the revenue** indicating that nearly half of the company's sales depend on a single product category.
- **Blouse is the most purchased item in clothing category**, while shirt is the least purchased item in clothing category
- **Gloves recorded the highest average review rating (3.86/4)**, while shirt had the lowest (3.64/4) indicating higher customer satisfaction with Gloves.
- Male customers generated **the highest revenue \$157,890**, contributing **68% of total revenue**, while female customers generated **the lowest revenue \$75,191 (32%)**.

- **Adult customers (26–45)** generated **\$87,576**, which is approximately 2.5 times the revenue generated by **Young Adults (18–25)**, who contributed **\$34,630**.
- Non-subscribers generated total revenue of **\$170,436** compared to **\$62,645** generated by subscribers. Non-subscribers accounted for **73%** of total revenue.
- The customer base consists of **3116 Loyal customers**, **701 Returning customers**, and **83 New customers**, with Loyal customers representing **80%** of all customers indicating strong customer retention.
- **Express shipping** recorded the highest average purchase amount of **\$60.48**, suggesting that customers choosing faster delivery tend to make higher-value purchases.

RECOMMENDATIONS

- Increase investment in the Clothing category by expanding product offerings, at the same time, develop targeted promotions for lower-performing categories, such as Outerwear, to diversify revenue sources and reduce dependence on a single category.
- Ensure sufficient inventory and promotional support for Blouses to meet customer demand. Conduct further analysis on Shirts to determine what is limiting sales and implement improvement strategies accordingly.
- Use customer feedback to identify improvement opportunities for lower-rated products, particularly Shirts.
- Design engagement campaigns aimed at increasing purchases among lower-contributing demographic groups while developing targeted marketing campaigns for the customer segments that contribute the most revenue.
- Boost Subscriptions – Promote exclusive benefits for subscribers.
- Rewarding repeat customers through personalized offers, loyalty points and exclusive promotions
- Encourage customers to upgrade to Express Shipping by offering free Express Shipping above a specified purchase threshold. This may increase average order values while improving customer experience.

TECHNOLOGIES USED

- Python – Data cleaning, preprocessing, feature engineering, and exploratory data analysis.
- PostgreSQL – Structured querying and business analysis.
- SQL – Data extraction, aggregation, customer segmentation, and analytical queries.
- Power BI – Interactive dashboard development and business reporting.
- Git & GitHub – Version control and project documentation.